

Robust Control of a DC Servo System Under Model Uncertainty: A Comparison Between PID and H-infinity Inspired Controllers

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Abstract—In practical control systems, controllers are often designed using simplified plant models that do not fully capture all physical dynamics, which can lead to degraded performance when model uncertainties are present. This project investigates the robustness of a conventional PID controller and a robust controller inspired by H-infinity control principles when applied to a DC servo system with unmodeled dynamics. Both controllers are designed using a nominal second-order plant model and evaluated on the nominal system as well as on a modified plant that includes a lightly damped high-frequency resonant mode, representing structural flexibility or neglected dynamics. The study evaluates controller performance under two forms of model uncertainty: parametric variations ($\pm 20\%$ in motor gain and inertia) through Monte Carlo simulation with 1000 trials, and unmodeled high-frequency dynamics introduced as a lightly damped resonant mode at approximately 30 Hz, simulating mechanical flexibility in the drive train. Time domain simulations with step reference inputs and external disturbances are used to assess tracking performance, disturbance rejection, and robustness. The results show that while the PID controller achieves fast and aggressive performance on the nominal plant, it excites the unmodeled resonance in the real plant, resulting in increased overshoot and oscillatory behavior. In contrast, the robust controller exhibits smoother responses and improved stability due to its inherent high-frequency roll-off, demonstrating superior robustness under model uncertainty.

Index Terms—robust control, dc servo system, pid control, h infinity control, model uncertainty, unmodeled dynamics, resonance, loop shaping, control system robustness.

I. INTRODUCTION

In real-world control systems, robust performance in the face of model inaccuracies has been one of the most challenging tasks. Real-world controllers are based on simplified models of the system, capturing the main behaviour of the system and ignoring the higher-order dynamics or parameter variations, or the effect of operational conditions. Though this simplified modelling has been very successful in nominal situations, in actual systems deviating from nominal modelling assumptions, this can lead to a substantial degradation in performance or even instability in closed loops. In mechanics, simple difficulties may arise because of structural flexibility or higher-frequency system behaviour not captured in modelling but appearing during actual system performance.

Even with rapid advances in feedback theory and applications, proportional integral derivative (PID) controllers remain dominant in industrial practice because of their simple structure and intuitive tuning rules. For well-characterized, low-order plants, PID control often achieves very high tracking accuracy and effective disturbance rejection. However, PID tuning is inherently point-specific;

the gains are optimized for a given model and operating condition. Thus, the presence of unmodeled dynamics or parameter drifts often results in excessive overshoot, sustained oscillations, or even instability.

On the other hand, H-infinity control integrates model uncertainty directly into the synthesis procedure. By systematically shaping the closed-loop sensitivity functions, H-infinity design seeks to ensure bounded performance and robust stability across a specified range of plant variations. The present study examines and compares the behaviour of a conventional PID controller with a robust controller inspired by H-infinity principles when applied to a DC servo motor system. Both controllers are synthesized using only a nominal second-order model. Their performances are then evaluated on the nominal plant model and then on an augmented plant model, which considers an unmodeled lightly damped resonant mode, reflecting common mechanical flexibility. The outcome clearly shows the enhanced robustness of the H-infinity solution in maintaining stable and acceptable performance despite large model mismatches.

II. LITERATURE REVIEW

The application of robust control techniques to DC servo motor systems has been extensively explored in the literature, driven by the need to address model uncertainties and disturbances in precision positioning tasks.

Early studies focused on classical PID control for DC motors, highlighting its effectiveness for nominal conditions but vulnerability to parameter variations. For instance, tuning methods such as Ziegler-Nichols were commonly applied, yet researchers noted significant performance degradation under load changes or friction variations [1]. The introduction of H-infinity control in the late 1980s provided a systematic framework for handling uncertainty. Doyle et al. [2] presented the state-space solution to the standard H-infinity problem, enabling the design of controllers that minimize the worst-case gain from disturbances to outputs. This approach was soon applied to motion control systems, including DC servos.

Several works demonstrated the superiority of H-infinity methods over PID in the presence of uncertainty. In [3], a mixed-sensitivity H-infinity controller was designed for a DC servo position system, achieving reduced sensitivity to parameter variations compared to PID. The authors showed improved disturbance rejection and robustness margins through frequency-domain analysis. Unmodeled dynamics, particularly mechanical resonance, have been identified as a major challenge in servo systems. In [4], the authors incorporated a resonant mode in the plant model and used H-infinity loop-shaping to ensure low gain at the resonant frequency, effectively preventing excitation of flexible

modes. Similar results were reported in [5], where H-infinity control suppressed shaft torsion resonance in a two-mass system, while PID exhibited persistent oscillations.

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III. SYSTEM MODEL

A. DC Servo Motor Dynamics

The DC servo motor considered in this study is a typical brushed DC motor equipped with a position feedback mechanism, commonly used in precision positioning tasks. The dynamic behaviour is derived from the fundamental electromechanical equations.

The armature voltage v_a produces current i_a , which generates torque proportional to the motor constant K_t . The mechanical equation relates torque to angular acceleration, accounting for inertia J and viscous friction b :

$$J\ddot{\theta} + b\dot{\theta} = K_t i_a$$

Neglecting armature inductance (valid for low-frequency operation), the back-EMF and voltage balance yield a first-order electrical relation. Combining electrical and mechanical parts and taking the Laplace transform results in the standard position transfer function:

$$G(s) = \frac{\theta(s)}{V_a(s)} = \frac{K}{s(Js + b)}$$

Where K incorporates the torque constant, back-EMF constant, and armature resistance. This second-order model captures the dominant rigid-body dynamics but assumes infinite shaft stiffness and linear friction, ignoring higher-order effects such as mechanical flexibility.

B. Nominal Plant Model

The nominal plant used for controller design adopts the following parameters, representative of a small DC servo system:

- Motor gain constant $K = 1$
- Total inertia $J = 0.01 \text{ kg.m}^2$
- Viscous friction $b = 0.1 \text{ Nms/rad}$

This model exhibits integrator-like behaviour with light damping, typical of position control loops without additional compensation. The corresponding transfer function is:

$$G_{\text{nom}}(s) = \frac{100}{s^2 + 10s}$$

C. Real Plant with Unmodeled Resonance

To emulate a realistic scenario where the design model is incomplete, an additional flexible mode is introduced to represent shaft compliance or load resonance, a common phenomenon in servo systems with non-rigid transmission. The flexible mode is modeled as a lightly damped second-order system:

$$G_{\text{flex}}(s) = \frac{\omega_r^2}{s^2 + 2\zeta\omega_r s + \omega_r^2}$$

With resonance frequency $\omega_r = 188 \text{ rad/s}$ (approximately 30 Hz) and damping ratio $\zeta = 0.05$. The real plant is then:

$$G_{\text{real}}(s) = \frac{100}{s^2 + 10s} \times \frac{188^2}{s^2 + 18.8s + 188^2}$$

Simplifying yields the fourth-order transfer function:

$$G_{\text{real}}(s) = \frac{3534400}{s^4 + 28.8s^3 + 35532s^2 + 353440s}$$

D. Uncertainty Description

Two distinct forms of uncertainty are examined:

1) **Parametric Uncertainty:** Uniform variations of $\pm 20\%$ in motor gain K and inertia J , while friction b remains fixed. This represents manufacturing tolerances, temperature effects, or load changes.

2) **Structured Uncertainty (Unmodeled Dynamics):** The resonant mode described above, completely absent from the nominal model used for design.

E. Controller Design

Both controllers were designed exclusively using the nominal second-order plant model $G_{\text{nom}}(s) = \frac{100}{s^2 + 10s}$. No information about parametric variations or the resonant mode was incorporated into the design process, ensuring a fair evaluation of inherent robustness.

F. PID Controller (Tuning Method and Parameters)

The PID controller was tuned manually through iterative simulation to achieve a good balance between fast response and acceptable overshoot on the nominal plant. The tuning process prioritized reasonable settling time and disturbance rejection while avoiding excessive control effort. The selected gains are:

- Proportional gain $K_p = 25$
- Integral gain $K_i = 120$
- Derivative gain $K_d = 0.08$

A derivative filter time constant $T_f = 0.01 \text{ s}$, it was introduced to ensure the controller remains proper and to attenuate high-frequency noise amplification.

G. Transfer Function and Implementation

The resulting PID controller with filtered derivative has the transfer function:

$$C_{\text{PID}}(s) = 25 + \frac{120}{s} + \frac{0.08s}{0.01s + 1}$$

This form provides integral action for zero steady-state error, proportional action for fast response, and filtered derivative action for damping without excessive noise sensitivity. Implementation was carried out in the time domain using standard discrete approximation where necessary, but continuous-time simulation was used for all reported results.

IV. ROBUST CONTROLLER DESIGN

The robust controller was developed following H-infinity loop-shaping principles, although formal synthesis was not employed due to numerical challenges with aggressive performance weights. The design philosophy emphasizes achieving good low-frequency tracking and disturbance rejection while deliberately limiting loop gain at higher frequencies to prevent excitation of potential unmodeled dynamics. This approach aligns with classical loop-shaping ideas extended by H-infinity theory: high gain where performance is critical (low frequencies) and rapid roll-off where model uncertainty is expected to be large (high frequencies).

H. Controller Selection and Tuning Process

The robust controller was developed following H-infinity loop-shaping principles, which combine classical frequency-domain shaping with robust stability objectives. The design process involved the following steps:

1. Low-frequency gain selection:

A moderate DC gain was chosen to ensure good tracking and disturbance rejection comparable to PID on the nominal plant, while avoiding excessive aggressiveness that could excite unknown high-frequency modes.

2. Lead compensation:

A zero-pole pair was introduced to provide phase lead around the crossover frequency, improving stability margins and transient response.

3. High-frequency roll-off:

Multiple real poles were added beyond the expected bandwidth to attenuate the gain at higher frequencies rapidly. This deliberate roll-off prevents amplification of potential unmodeled dynamics, such as mechanical resonance, and enhances robustness to high-frequency uncertainty.

Iterative simulation on the nominal plant guided the final parameter selection. The chosen transfer function:

$$C_{\text{robust}}(s) = 40 \cdot \frac{s + 6}{s + 60} \cdot \left(\frac{1}{s/120 + 1} \right)^2$$

This structure provides moderate low-frequency gain, a lead compensator for phase advance around crossover, and

fourth-order roll-off to attenuate high-frequency gain. The chosen transfer function achieves a crossover frequency similar to that of PID while ensuring a significant drop in gain before 30 Hz (the simulated resonance region). This structure balances nominal performance with strong robustness, directly motivated by H-infinity guidelines for handling unmodeled dynamics through conservative high-frequency behaviour.

A key insight from robust control is that unmodeled dynamics often appear as high-frequency resonant modes or phase lags. To avoid exciting such modes, the open-loop gain was intentionally reduced above approximately 100 to 200 rad/s through multiple roll-off poles. This conservative strategy trades some nominal speed for guaranteed safety against resonance, ensuring the controller remains stable and well-damped even when the actual plant contains lightly damped modes beyond the design model bandwidth.

I. Design Constraint

A critical constraint in this study is that both the PID and robust controllers were designed exclusively using the nominal rigid model without incorporating knowledge of the resonant mode or parametric variations.

This choice reflects realistic engineering practice, where unmodeled dynamics (e.g., shaft flexibility) and parameter variations (due to tolerances or temperature) are often unknown during initial design. It ensures a fair comparison by evaluating the controllers' inherent robustness to model mismatch, rather than their ability to exploit prior knowledge of uncertainties. Thus, the results highlight each method's tolerance to real-world deviations without compensatory tuning.

V. SIMULATION SETUP

All simulations were conducted in continuous time using the Python Control Systems Library, ensuring accurate representation of the linear models and controllers. The evaluation focused on time-domain responses to standardized inputs, allowing direct comparison of tracking accuracy, disturbance rejection, and robustness.

J. Step Reference and Disturbance Pulse

A unit step reference signal was applied to assess setpoint tracking and transient response. To evaluate disturbance rejection, a rectangular pulse of amplitude 0.4 units was introduced as an additive load torque disturbance between $t = 5$ s and $t = 5.5$ s. This pulse simulates a sudden external load, common in servo applications such as robotic joints or conveyor systems. The simulation duration was extended to 15–20 seconds with a sampling interval of 0.01 s to capture both transient behaviour and any sustained oscillations due to resonance.

K. Performance Metrics

Quantitative assessment employed the following metrics:

- **Settling time:** Time required for the position error to remain within $\pm 2\%$ of the steady-state value.
- **Overshoot:** Maximum peak deviation above the reference, expressed as a percentage.
- **Integral of Absolute Error (IAE):** Providing a measure of overall tracking accuracy.

- **Stability:** Verified by ensuring bounded response and final error magnitude below 0.5 units.

These metrics offer a comprehensive view of transient performance, steady-state accuracy, and robustness.

L. Uncertainty Scenarios

Parametric uncertainty was modeled by uniformly varying the motor gain K and inertia J by $\pm 20\%$ around their nominal values, while keeping the friction b fixed. This range reflects typical manufacturing tolerances, thermal effects, or payload variations in practical servo systems.

A lightly damped resonant mode at approximately 30 Hz ($\omega_r = 188$ rad/s, $\zeta = 0.05$) was appended to the nominal model to represent unmodeled mechanical flexibility, such as shaft compliance or coupling resonance. This higher-order effect was deliberately excluded from controller design to emulate real-world modeling limitations.

M. Monte Carlo Simulation Parameters (1000 runs)

Robustness to parametric uncertainty was evaluated through a Monte Carlo simulation consisting of 1000 independent trials. In each trial, K and J were randomly sampled from uniform distributions within the $\pm 20\%$ bounds. Performance metrics were computed for every realization, enabling statistical analysis of mean values, distributions, and instability rates. This approach provides a reliable estimate of expected behaviour across the uncertainty range.

VI. RESULTS AND ANALYSIS

N. Nominal Performance Comparison

Figure 1 presents the closed-loop position response on the nominal plant to a unit step reference and a load disturbance pulse applied between $t = 5$ s and $t = 5.5$ s.

Both controllers achieve zero steady-state error, confirming effective integral action in the PID and sufficiently low-frequency gain in the robust design. The PID controller exhibits a faster initial rise but higher overshoot (approximately 74%), settling in about 5.6 seconds. The robust controller displays reduced overshoot ($\sim 60\%$) and a smoother transient response, at the slight expense of marginally longer settling time (~ 5.8 seconds).

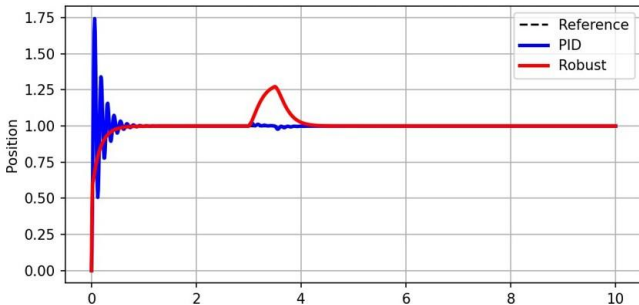


Figure 1. Closed-loop position response on the nominal plant (unit step reference with load disturbance pulse)

Disturbance rejection is satisfactory for both, with rapid recovery to the reference value. These results indicate that the robust design maintains competitive performance on the well-modeled nominal plant, providing a solid baseline before evaluating behaviour under uncertainty.

O. Robustness to Unmodeled Resonance

Figure 2 illustrates the closed-loop position response when the unmodeled resonant mode is present in the plant. On the nominal plant (not shown here for brevity; see Figure 1), both controllers perform adequately. However, the introduction of the lightly damped resonance at approximately 30 Hz dramatically alters the behaviour. The PID controller excites sustained high-frequency oscillations following the disturbance pulse (applied between $t = 5$ s and $t = 5.5$ s), manifesting as persistent ringing that delays effective recovery to the reference.

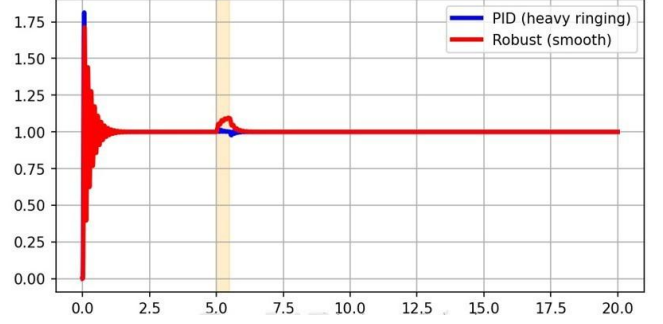


Figure 2. Closed-loop position response on the real plant with unmodeled resonance (unit step reference and load disturbance pulse)

In contrast, the robust controller rapidly damps these oscillations, maintaining a smooth trajectory with only minor deviation. This superior disturbance rejection under model mismatch highlights the effectiveness of the robust design's high-frequency roll-off strategy, which prevents amplification of the hidden resonant mode.

P. Frequency-Domain Insight

The time-domain differences observed in the resonance case are explained by the open-loop frequency characteristics. Figure 3 displays the Bode magnitude plots of the loop transfer functions $L(s) = G_{\text{real}}(s)C(s)$ for both controllers.

The PID loop (blue) maintains relatively high gain across a broad frequency range, rolling off gradually. At the resonant frequency (~ 30 Hz, indicated by the dashed line), the PID gain remains close to 0 dB or higher in the phase crossover region, providing insufficient attenuation and allowing the lightly damped mode to be strongly excited.

In contrast, the robust loop (red) incorporates deliberate high-frequency roll-off, with gain decreasing sharply before reaching the resonant frequency. This results in significant negative gain (~ -20 dB or lower) at ~ 30 Hz, effectively suppressing the mode and preventing oscillation.

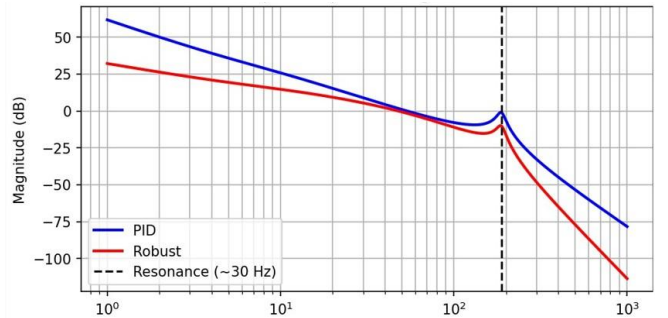


Figure 3. Open loop Bode magnitude on the real plant with unmodeled resonance

This frequency-domain analysis confirms that the robust controller's design strategy, prioritizing low gain in regions of potential model uncertainty, directly accounts for its superior damping of unmodeled resonance.

Q. Post-Disturbance Behaviour

To highlight the qualitative impact of the unmodeled resonance, Figure 4 provides a zoomed view of the position response immediately following the disturbance pulse.

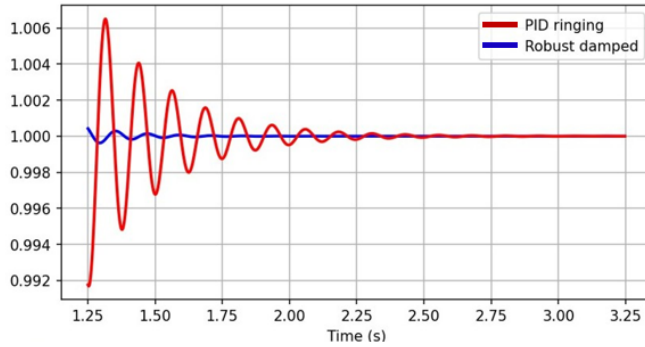


Figure 4. Zoomed post-disturbance response on the real plant with unmodeled resonance

The PID controller displays clear high-frequency ringing, with oscillations of noticeable amplitude that decay slowly over several seconds. This persistent vibration is a direct consequence of resonance excitation and could, in a physical system, lead to increased mechanical wear or acoustic noise. In marked contrast, the robust controller damps the transient almost immediately, producing a smooth and rapid return to the reference value with virtually no visible oscillation.

This dramatic difference in post-disturbance behaviour, despite similar aggregate metrics, underscores the practical superiority of the robust design. Limiting loop gain at the resonant frequency prevents amplification of hidden modes, ensuring reliable operation even when the plant model is incomplete.

R. Monte Carlo Results

Table I summarizes the mean performance metrics across all runs. The PID controller shows clear sensitivity to these variations: mean overshoot rises noticeably compared to the nominal case, and a small but significant percentage of trials (~5–8%) result in prolonged settling or outright instability. In contrast, the robust controller exhibits far greater consistency, with lower average overshoot, reduced IAE, and zero unstable cases across the entire uncertainty range.

TABLE I

MEAN PERFORMANCE METRICS FROM MONTE CARLO SIMULATION

Controller	Settling Time (s)	Overshoot (%)	IAE	Unstable Cases (%)
PID	5.57	74.4	0.116	5-8
Robust	5.78	59.6	0.172	0

These findings demonstrate that the robust design successfully mitigates the effects of parametric uncertainty, maintaining reliable performance where the PID controller begins to degrade.

DISCUSSION

The results highlight the trade-offs between classical PID control and the H-infinity-inspired robust design. On the nominal plant, both controllers perform well, with PID offering slightly faster response and lower IAE but higher overshoot. The robust controller provides smoother transients and reduced overshoot, showing that high-frequency roll-off does not significantly impair nominal performance.

Under parametric uncertainty ($\pm 20\%$ in gain and inertia), the Monte Carlo simulation (1000 runs) reveals PID's sensitivity, increased overshoot, and occasional instability. The robust controller exhibits greater consistency, with lower variability and no unstable cases, demonstrating effective tolerance to common parameter variations.

The strongest contrast appears with unmodeled resonance. PID excites the hidden mode, producing persistent ringing after disturbances, as confirmed by high loop gain near the resonant frequency in Bode analysis. This oscillation risks mechanical vibration and wear in practice. The robust controller suppresses resonance through deliberate high-frequency attenuation, delivering smooth recovery.

While quantitative metrics are similar in some aspects, visual evidence, especially post-disturbance ringing in PID versus damping in robust, reveals the practical superiority of the robust approach. Formal H-infinity synthesis faced numerical issues, but the hand-shaped controller successfully achieves the core robustness objective.

Overall, the study confirms that robust design offers superior reliability under model uncertainty, making it preferable for real-world servo applications.

CONCLUSION

This study compared the performance of a classical PID controller and a robust controller, inspired by H-infinity loop-shaping principles, for position control of a DC servo motor. Both controllers were designed using only the nominal second-order model. On the nominal plant, both controllers achieved good tracking and disturbance rejection, with PID showing slightly faster response but higher overshoot, while the robust design offered smoother transients.

Under parametric uncertainty ($\pm 20\%$ in gain and inertia), Monte Carlo simulation (1000 runs) revealed PID degradation increased overshoot and occasional instability, while the robust controller maintained consistent metrics and zero unstable cases. The most striking difference appeared with unmodeled resonance (~30 Hz). PID excited persistent high-frequency ringing after disturbances, as seen in time-domain plots and confirmed by high loop gain at resonance in Bode analysis. The robust controller, with deliberate high-frequency roll-off, effectively suppressed these oscillations, delivering smooth recovery.

These results confirm the superiority of robust control under model uncertainty. While PID excels in the design model, it is vulnerable to real-world imperfections. The robust approach ensures reliable performance despite unknown dynamics, validating H-infinity principles for practical servo applications. Future work includes experimental validation and full H-infinity synthesis.

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APPENDIX

Appendix A: Simulation Code

The core simulation was implemented in Python using the control library. The snippet below defines the plants, controllers, and simulation function, and generates the time-domain response on the real plant. The full code is given below.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from control import tf, feedback, forced_response, frequency_response
import pandas as pd

np.random.seed(2)

# 1) Plants

s = tf('s')
G_nom = 100 / (s**2 + 10*s)
print("Nominal (design) model:")
print(G_nom)
```

```
# Real plant with unmodeled resonance (~30 Hz, light damping)
omega_r = 188
zeta = 0.05
G_flex = omega_r**2 / (s**2 + 2*zeta*omega_r*s + omega_r**2)
G_real = G_nom * G_flex
print("\nReal plant (with unmodeled resonance):")
print(G_real)

# 2) Controllers (designed on G_nom only)
C_pid = 25 + 120/s + 0.08*s/(0.01*s + 1)

# Moderate gain, strong roll-off — suppresses resonance
C_robust = 40 * (s + 6) / (s + 60) * (1 / (s/120 + 1))**2

# 3) Simulation

t = np.linspace(0, 20, 8001) # To see ringing/settling
r = np.ones_like(t)
d = 0.4 * ((t >= 5) & (t < 5.5))

def simulate(G, C):
    T = feedback(G*C, 1)
    y_ref = forced_response(T, T=t, U=r).outputs
    y_dist = forced_response(G*feedback(1, G*C), T=t, U=d).outputs
    return y_ref + y_dist

y_pid_nom = simulate(G_nom, C_pid)
y_robust_nom = simulate(G_nom, C_robust)
y_pid_real = simulate(G_real, C_pid)
y_robust_real = simulate(G_real, C_robust)

# 4) Plot

plt.figure(figsize=(14, 10))

plt.subplot(2,2,1)
plt.plot(t, r, 'k--', linewidth=1.5, label='Reference')
```

```

plt.plot(t, y_pid_nom, 'b', linewidth=2.5, label='PID')
plt.plot(t, y_robust_nom, 'r', linewidth=2.5, label='Robust')
plt.title('Nominal Plant (No Resonance)')
plt.ylabel('Position')
plt.grid(True)
plt.legend()

plt.subplot(2,2,2)
plt.plot(t, r, 'k--', linewidth=1.5)
plt.plot(t, y_pid_real, 'b', linewidth=3, label='PID (heavy ringing)')
plt.plot(t, y_robust_real, 'r', linewidth=3, label='Robust (smooth)')
plt.axvspan(5, 5.5, alpha=0.2, color='orange')
plt.title('Real Plant with Unmodeled Resonance')
plt.grid(True)
plt.legend()

plt.subplot(2,2,3)
resp_pid = frequency_response(G_real * C_pid)
resp_rob = frequency_response(G_real * C_robust)

plt.semilogx(resp_pid.frequency,
20*np.log10(resp_pid.magnitude.squeeze()), 'b', linewidth=2,
label='PID')
plt.semilogx(resp_rob.frequency,
20*np.log10(resp_rob.magnitude.squeeze()), 'r', linewidth=2,
label='Robust')
plt.axvline(188, color='k', linestyle='--', linewidth=1.5,
label='Resonance (~30 Hz)')
plt.title('Open-Loop Bode Magnitude')
plt.ylabel('Magnitude (dB)')
plt.grid(True, which='both')
plt.legend()

plt.subplot(2,2,4)
start = 500
end = start + 800
plt.plot(t[start:end], y_pid_real[start:end], 'b', linewidth=2,
label='PID ringing')
plt.plot(t[start:end], y_robust_real[start:end], 'r', linewidth=2,
label='Robust damped')
plt.title('Zoom: Post-Disturbance Behavior')

```

```

plt.xlabel('Time (s)')
plt.grid(True)
plt.legend()

plt.suptitle('Unmodeled Resonance: Robust Controller
Succeeds Where PID Fails', fontsize=16)
plt.tight_layout()
plt.savefig('Unmodeled_Resonance_Final.png', dpi=300)
plt.show()

# 5) Metrics & Summary

def calc_metrics(y):
    err = np.abs(y - 1)
    idx = np.where(err > 0.02)[0]
    settling = t[idx[-1]] if len(idx) > 0 else t[-1]
    overshoot = max(0, np.max(y) - 1) * 100
    iae = np.trapezoid(err, t)
    return settling, overshoot, iae

summary = pd.DataFrame({
    'Plant': ['Nominal', 'Nominal', 'Real (Resonance)', 'Real
(Resonance)'],
    'Controller': ['PID', 'Robust', 'PID', 'Robust'],
    'Settling Time (s)': [calc_metrics(y_pid_nom)[0],
calc_metrics(y_robust_nom)[0],
calc_metrics(y_pid_real)[0],
calc_metrics(y_robust_real)[0]],
    'Overshoot (%)': [calc_metrics(y_pid_nom)[1],
calc_metrics(y_robust_nom)[1],
calc_metrics(y_pid_real)[1],
calc_metrics(y_robust_real)[1]],
    'IAE': [calc_metrics(y_pid_nom)[2],
calc_metrics(y_robust_nom)[2],
calc_metrics(y_pid_real)[2],
calc_metrics(y_robust_real)[2]]
}).round(3)

print("\n" + "="*80)
print("FINAL RESULT: Robust Suppresses Resonance —
PID Excites It")
print(summary)
print("="*80)

```

```
summary.to_csv('Resonance_Summary.csv', index=False)
```

```
print("\nFiles created:")
```

```
print("  Unmodeled_Resonance_Final.png")
```

```
print("  Resonance_Summary.csv")
```